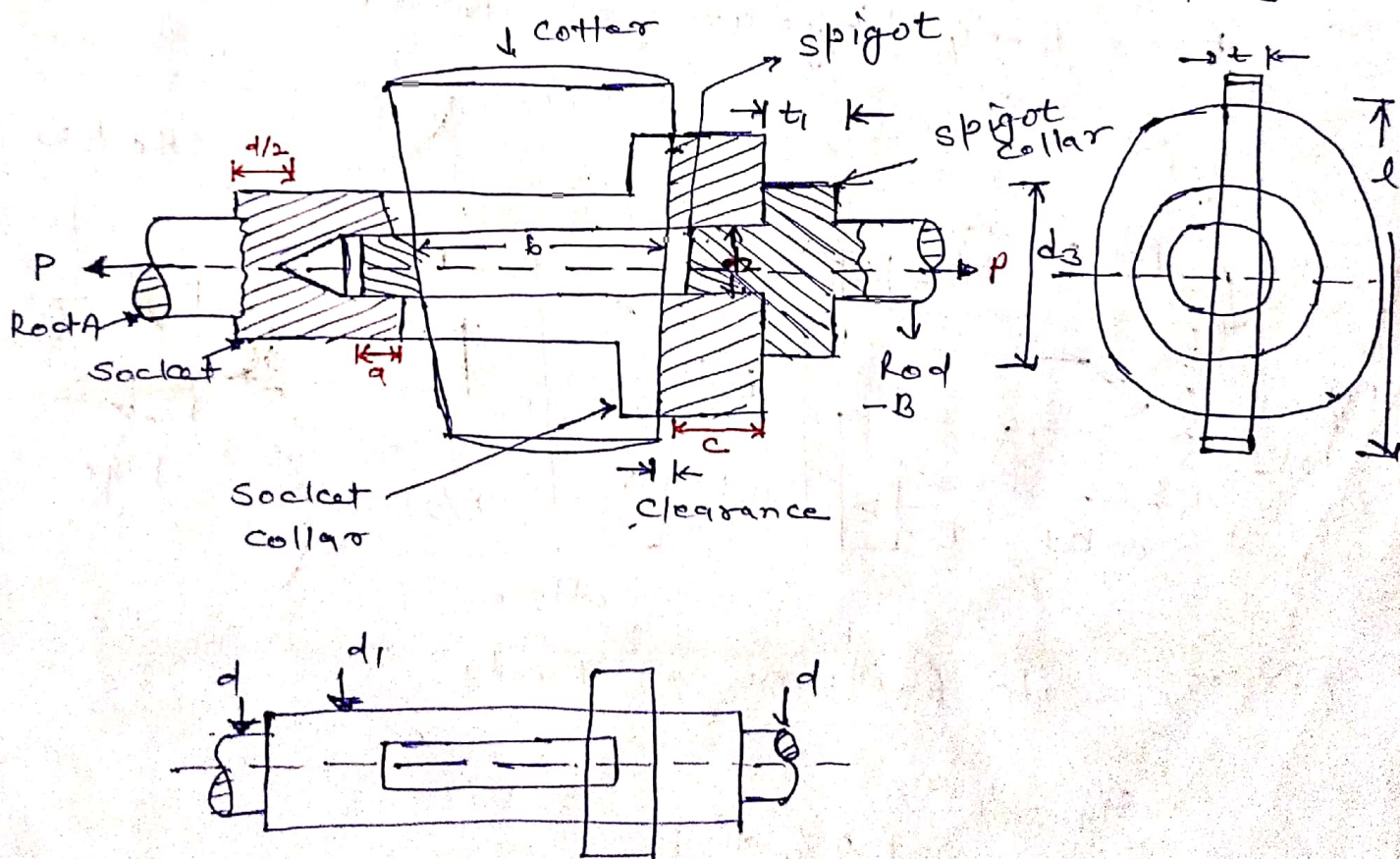


Design of Machine elements subjected to direct stress.

Cotter Pin Joint!

It is used to connect two co-axial rods, which are subjected to either axial tensile force or axial compressive force. It is also used to connect a rod on one side with some machine parts like a crosshead or base plate on the other side. It is not used for connecting shafts that rotate and transmit torque. The principle of wedge action is used in a cotter joint.

Amount by which the two rods are drawn together is called the draw of the cotter. It mainly consists of three parts socket, cotter and spigot.



P = Tensile force acting on rods (N)

d = dia. of each rod (mm)

d_1 = outside dia. of socket (mm)

d_2 = dia. of spigot or inside dia. of socket (mm)

d_3 = dia. of spigot-collar (mm)

d_4 = dia. of socket-collar (mm)

a = distance from end of slot to the end of spigot on rod-B (mm)

b = Mean width of cotter (mm)

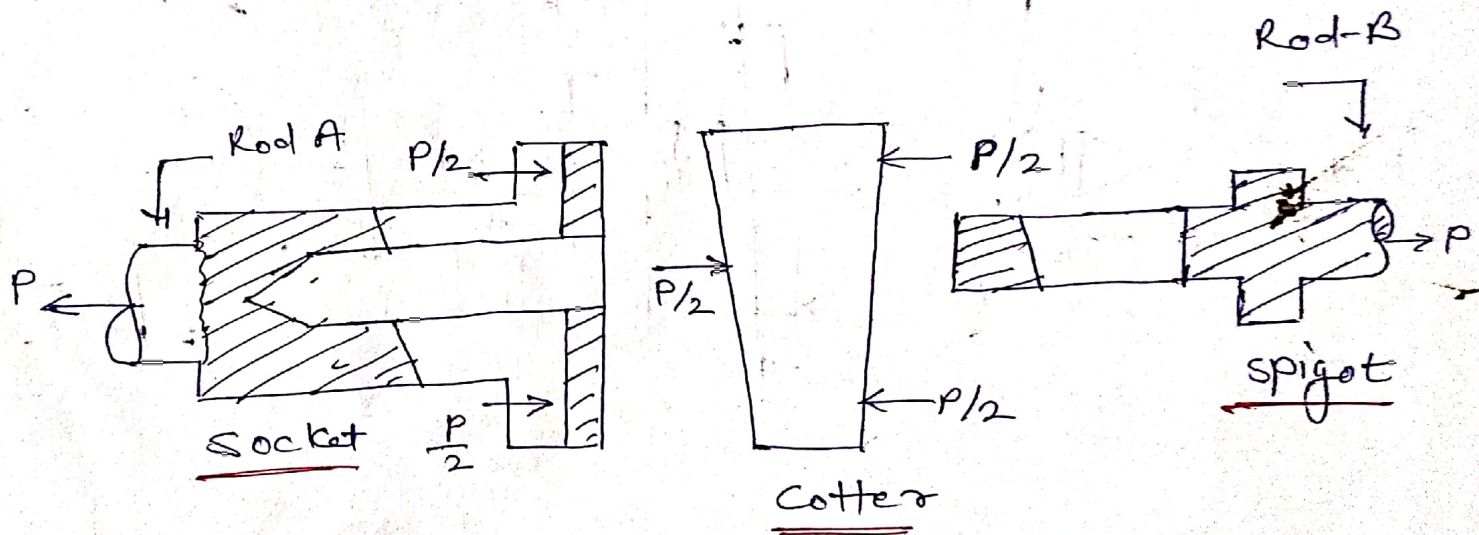
c = axial distance from slot to end of socket collar (mm)

t = thickness of cotter (mm)

t_1 = thickness of spigot-collar (mm)

l = length of cotter (mm)

Free Body diagram of forces.



(1) Tensile failure of Rods

$$\sigma_t = \frac{P}{\left[\frac{\pi}{4} d^2 \right]}$$

$$d = \sqrt{\frac{4P}{\pi \sigma_t}} \quad \text{--- (1)}$$

σ_t : Permissible tensile stress for rods

(3)

(2) Tensile failure of spigot

$$\text{Area of section at } xx = \left[\frac{\pi}{4} d_2^2 - d_2 t \right]$$

$$\sigma_t = \frac{P}{\left(\frac{\pi}{4} d_2^2 - d_2 t \right)}$$

$$P = \left(\frac{\pi}{4} d_2^2 - d_2 t \right) \sigma_t \quad \text{--- (2)}$$

(find d_2)

$$t = 0.31 d \quad \text{--- (3)}$$

(3)

Tensile failure of socket

$$\text{Area} = \left[\frac{\pi}{4} (d_1^2 - d_2^2) - (d_1 - d_2) t \right]$$

$$\sigma_t = P / \text{area}$$

$$\sigma_t = \frac{P}{\left[\frac{\pi}{4} (d_1^2 - d_2^2) - (d_1 - d_2) t \right]}$$

$$P = \left[\frac{\pi}{4} (d_1^2 - d_2^2) - (d_1 - d_2) t \right] \sigma_t \quad \text{--- (4)}$$

(find d_1)

(4)

Shear failure of spigot end

$$\tau = \frac{P}{2(a d_2)}$$

$$P = 2 a d_2 \tau \quad \text{--- (5)}$$

(find a)

τ : Permissible shear stress for the spigot.

(5)

Shear failure of socket end

$$\text{Area} = (d_1 - d_2) \times c$$

$$\tau = P / 2(d_1 - d_2) \times c$$

(4)

$$P = 2(d_4 - d_2)ct - (6)$$

Find out c

(6)

Shear Failure of Cotter

$$c = \frac{P}{2(bt)}$$

$$P = 2btz - (7)$$

(Find out b)

(7)

Crushing failure at spigot end

$$\sigma_c = \frac{P}{td_2} - (8)$$

(8)

Crushing failure of socket end

$$\sigma_c = \frac{P}{(d_4 - d_2)t} - (9)$$

(9)

Bending failure of cotter.

$$x = \frac{1}{3}y$$

$$x = \frac{1}{3} \left(\frac{d_4 - d_2}{2} \right)$$

$$x = \frac{(d_4 - d_2)}{6} - (10)$$

$$M_b = P/2 \left[\frac{d_2}{2} + x \right] - P/2 (z)$$

$$= P/2 \left[\frac{d_2}{2} + \frac{d_4 - d_2}{6} \right] - \frac{P}{2} \left[\frac{d_2}{4} \right]$$

$$= \frac{P}{2} \left[\frac{d_2}{4} + \frac{d_4 - d_2}{6} \right]$$

$$I = \frac{t b^3}{12} \quad , \quad y = b/2$$

$$\sigma_b = \frac{M y}{I}$$

$$\sigma_b = \frac{P/2 \left[\frac{d_2}{4} + \frac{d_2 y}{6} \right] b/2}{\left(\frac{t b^3}{12} \right)} \quad \text{--- (11)}$$

Standard

$$d_1 = 1.75d$$

$$d_2 = 1.21d$$

$$d_3 = 1.5d$$

$$d_4 = 2.4d$$

$$a = c = 0.75d$$

$$b = 1.6d$$

$$t_1 = 0.45d$$

clearance 1-3 to 3mm

Table for cotter = 1 in 32

Design Procedure for Cotter joint!

(1) Calculate the diameter of each rod.

$$d = \sqrt{\frac{4P}{\pi \sigma_t}} \quad \text{--- (1)}$$

(2) calculate the thickness of the cotter by

$$t = 0.31d \quad \text{--- (2)}$$

(3) Calculate the diameter d_2 of the spigot on the basis of tensile stress

$$P = \left[\frac{\pi}{4} d_2^2 - d_2 t \right] \sigma_t \quad \text{--- (3)}$$

(4) Calculate the outside diameter d_1 of the socket on the basis of tensile stress in the socket.

$$P = \left[\frac{\pi}{4} (d_1^2 - d_2^2) - (d_1 - d_2)t \right] \sigma_c \quad \text{--- (4)}$$

- (5) The diameter of the spigot collar d_3 and the socket collar d_4 are calculated by

$$\begin{cases} d_3 = 1.5d \\ d_4 = 2.4d \end{cases} \quad \text{--- (5)}$$

- (6) The dimension a & c are calculated by the

$$a = c = 0.75d \quad \text{--- (6)}$$

- (7) Calculate the width b of the cotter by shear and bending consideration and select width whichever is maximum between these two values.

$$b = \frac{P}{2\tau t} \quad \text{--- (7)}$$

$$b = \frac{3P}{t\sigma_b} \left[\frac{d_2}{4} + \frac{d_4 - d_2}{6} \right] \quad \text{--- (8)}$$

- (8) Check the crushing & shear stress in the spigot by

$$\sigma_c = \frac{P}{td_2} \quad \text{--- (9)}$$

$$\tau = \frac{P}{2ad_2} \quad \text{--- (10)}$$

- ⑨ check the crushing & shear stresses in the socket end

$$\sigma_c = \frac{P}{(d_4 - d_2) t} \quad \text{--- (11)}$$

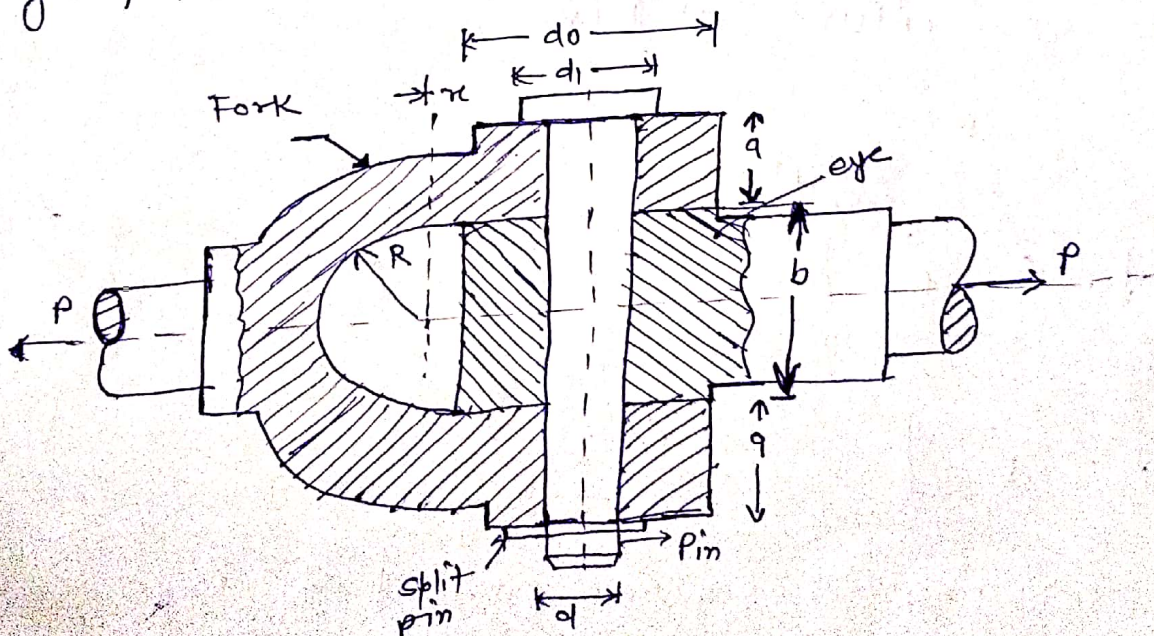
$$\tau = \frac{P}{2(d_4 - d_2) c} \quad \text{--- (12)}$$

- ⑩ Calculate the thickness t_1 of the spigot collar by the following relationship.

$$t_1 = 0.45 d \quad \text{--- (13)}$$

Design of ~~Spigot and Socket~~ Knuckle Joint

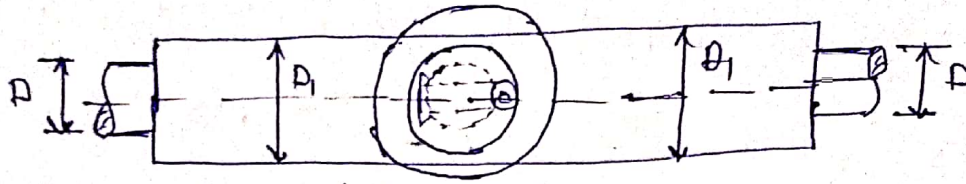
It is used to connect two rods whose axes either coincide or intersect and lie in one plane. It used to transmit axial tensile force. It popular in machines and structure. It consists of three parts. namely eye, fork & pin.



8

Rod A

Rod-B



D = dia. of each rod (mm)

D_1 = Enlarged dia. of each rod

d = Dia. of knuckle pin

d_0 = outside dia. of eye of fork

a = thickness of each eye of fork

b = " " eye end of rod-B (mm)

d_1 = diameter of pin head (mm)

r = distance of the centre of fork neck from the eye (mm).

① Tensile failure of rods.

$$\sigma_t = \frac{P}{\left(\frac{\pi}{4} D^2\right)}$$

$$D = \sqrt{\frac{4P}{\pi \sigma_t}} \quad \text{--- (1)}$$

$$D_1 = 1.1 D \quad \text{--- (2)}$$

② Shear failure of pin

$$\tau = \frac{P}{2\left(\frac{\pi}{4} d^2\right)}$$

$$d = \sqrt{\frac{2P}{\pi \tau}} \quad \text{--- (3)}$$

$$d = D \quad \text{--- (4)}$$

③ Crushing failure of pin in eye

$$\sigma_c = \frac{P}{2 \times a} \quad \text{--- (5)}$$

$$\sigma_c = \frac{P}{b \times a} \quad \text{--- (6)}$$

(4) Crushing Failure of Pin in fork

$$\sigma_c = \frac{P}{2 \times a} \quad \text{--- (7)}$$

(5) Bending failure of Pin!

$$x = \frac{1}{3} a \quad \text{also } z = \frac{1}{2} \left(\frac{1}{2} b \right)$$

$$z = \frac{1}{4} b$$

$$M_b = \frac{P}{2} \left[b/2 + x \right] - \frac{P}{2} (z)$$

$$= \frac{P}{2} \left[b/2 + \frac{a}{3} \right] - \frac{P}{2} \left[b/4 \right]$$

$$M_b = \frac{P}{2} \left[\frac{b}{4} + \frac{a}{3} \right]$$

$$\text{Also } I = \frac{\pi d^4}{64}, \quad y = d/2$$

$$\sigma_b = \frac{M_b y}{I}$$

$$\sigma_b = \frac{\frac{P}{2} \left[\frac{b}{4} + \frac{a}{3} \right] d/2}{\frac{\pi d^4}{64}}$$

(10)

$$\sigma_b = \frac{32}{\pi d^3} \times \frac{P}{2} \left[\frac{b}{4} + \frac{a}{3} \right] \quad \text{--- (8)}$$

(6) Tensile failure of eye!

$$\text{Area} = b(d_o - d)$$

$$\sigma_t = P/A$$

$$\sigma_t = \frac{P}{b(d_o - d)} \quad \text{--- (9)}$$

(7) Shear failure of eye!

$$\tau = \frac{P}{2 \left[\frac{b(d_o - d)}{2} \right]}$$

$$\tau = \frac{P}{b(d_o - d)} \quad \text{--- (10)}$$

$$d_o = 2d \quad \text{--- (11)}$$

(8) Tensile failure of fork!

$$\text{Area} = 2a(d_o - d)$$

$$\sigma_t = \frac{P}{2a(d_o - d)} \quad \text{--- (12)}$$

(9) Shear failure of fork!

$$\tau = \frac{P}{2a(d_o - d)} \quad \text{--- (13)}$$

$$a = 0.75 D \quad \text{--- (14)}$$

$$b = 1.25 D \quad \text{--- (15)}$$

$$d_1 = 1.5d$$

— (16)

$$\boxed{\sigma_c = 10 \text{ mm}}$$

Procedure for design!

- ① Calculate dia. of each rod

$$D = \sqrt{\frac{4P}{\pi \sigma_t}} \quad \text{— (1)}$$

- ② Calculate enlarged dia. of each rod by empirical relationship

$$\boxed{D_1 = 1.1 D} \quad \text{— (2)}$$

- ③ Calculate dimension a & b by

$$a = 0.75 D$$

$$b = 1.25 D$$

- ④ Calculate diameters of the pin by shear

$$d = \sqrt{\frac{2P}{\pi \tau}}$$

$$d = \sqrt[3]{\frac{32}{\pi \sigma_b} \times \frac{P}{2} \left(\frac{b}{4} + \frac{a}{3} \right)}$$

(whichever is max^m)

- ⑤ Calculate dimension d_0 & d_1 by

$$d_0 = 2d$$

$$d_1 = 1.5d$$

- ⑥ check the tensile, crushing & shear stresses in the eye.

$$\sigma_t = \frac{P}{b(d_o - d)}$$

$$\sigma_c = \frac{P}{bd}$$

$$\tau = \frac{P}{b(d_o - d)}$$

check the tensile, crushing & shear in bolts

(7)

$$\sigma_t = \frac{P}{2a(d_o - d)}$$

$$\sigma_c = \frac{P}{2ad}$$

$$\tau = \frac{P}{2a(d_o - d)}$$

Design of screw fastening :-